

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What is the **exact depth** that separates an erosion from an ulcer?
2. Which chronic gastritis causes **pernicious anaemia**, and why does the other one not?
3. Are **duodenal** or **gastric** ulcers more common, and by how much?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Acute disease — depth is everything

	Acute gastritis	Acute peptic ulceration
Depth of loss	Mucosal inflammation; surface epithelium intact . Erosion (loss above muscularis mucosa) only in severe cases.	Tissue loss extends BELOW the muscularis mucosa
Nature	Transient	Acute but deeper
Findings	Oedema and congestion of mucosa	Rounded, usually <1 cm
Symptoms	May be asymptomatic; or epigastric pain, nausea, vomiting. Severe: ulceration, haemorrhage, haematemesis, melaena, rarely massive blood loss.	Same spectrum, greater bleeding risk

The three named acute ulcers — learn the trigger

Name	Setting
Stress ulcer	Shock, sepsis , or severe trauma
Curling ulcer	Severe burns
Cushing ulcer	Intracranial disease

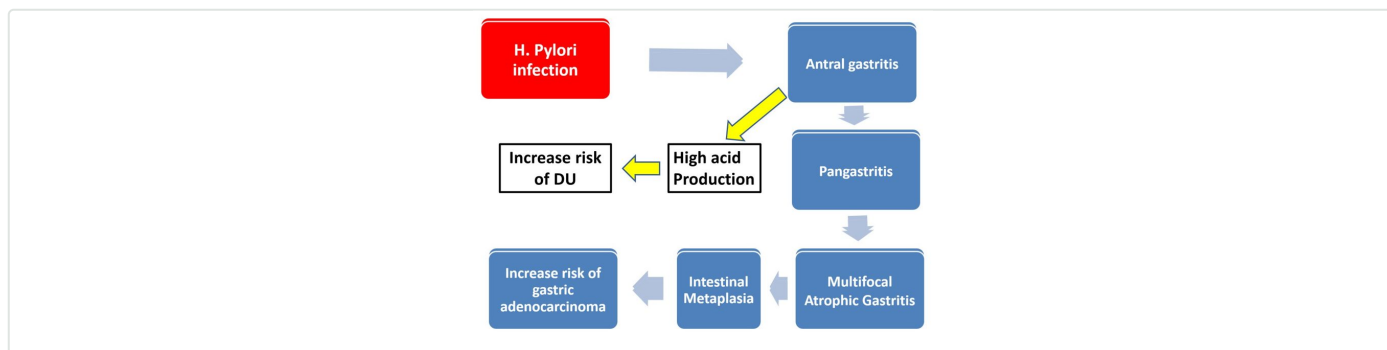
Also caused by **NSAID therapy**. Mnemonic that works: **C**urling → **C**ooking (burns); **C**ushing → **C**ranium.

2 Chronic gastritis — H. pylori vs autoimmune

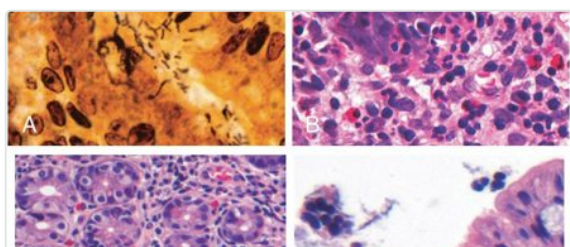
Three major types: **H. pylori-induced** (the most common form), **autoimmune**, and **chemical (reactive)**. Chronic gastritis gives **less severe but more persistent** symptoms than the acute form.

Feature	H. pylori gastritis	Autoimmune gastritis
Frequency	Most common form overall	<10% of chronic gastritis — but the commonest cause of gastric atrophy , and the commonest form in patients without H. pylori
Mechanism	Bacterial infection	Antibodies against parietal cells → loss of the cells that secrete HCl and intrinsic factor
Acid	Can cause hypochlorhydria	Achlorhydria — parietal cell damage is far more severe
B12	Not associated with pernicious anaemia	No intrinsic factor → ileal B12 absorption fails → megaloblastic (pernicious) anaemia
Histology	Spiral bacilli on Warthin-Starry silver stain ; intraepithelial and lamina propria neutrophils; lymphoid aggregates	Gastric atrophy — endoscopically whitish discoloration with visible vessels
Leads to	PUD · gastric adenocarcinoma · gastric MALToma (NHL)	Gastric atrophy · pernicious anaemia · carcinoma risk

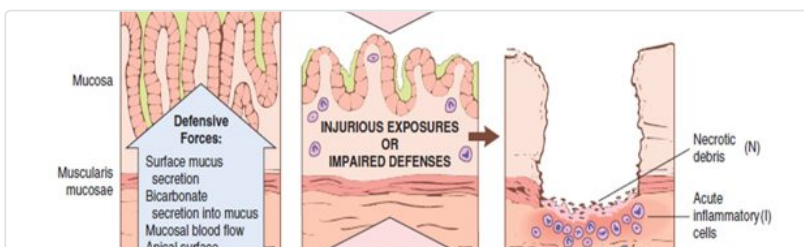
The exam's favourite distinction: H. pylori can lower acid, but only autoimmune gastritis produces **achlorhydria and pernicious anaemia**, because only it destroys parietal cells severely enough.



H. pylori → antral gastritis → pangastritis → multifocal atrophic gastritis → intestinal metaplasia → increased risk of gastric adenocarcinoma. The antral route instead raises acid and the risk of DU.



H. pylori — silver stain, neutrophils, lymphoid aggregates



Damaging forces vs mucosal defences — the imbalance behind gastritis and PUD

3 Peptic ulcer disease

Definition	Chronic lesions occurring anywhere in the GIT exposed to acidic peptic juices .
Sites	Stomach (GU) · duodenum (DU) · lower Barrett esophagus · Meckel diverticulum with gastric heterotopia .
Pathogenesis	The same imbalance of mucosal defences vs damaging forces that causes chronic gastritis: H. pylori , aspirin and NSAIDs , cigarette smoking , increased HCl and pepsin secretion.
Morphology	Solitary in >80%. Round to oval, sharply punched-out . Regular margins with radiating mucosal folds . Base smooth and clean from peptic digestion of exudate.

DU vs GU

	Duodenal ulcer	Gastric ulcer
Frequency	Four times more common	Less common
Site	Within a few cm of the pyloric valve , anterior duodenal wall	Near the interface of body and antrum , on the lesser curvature
H. pylori	Present in almost all patients	Present in the majority
Malignant change	Less likely	More likely (still very rare)

Histology of PUD — **four layers, always in this order from the lumen down:** **Necrotic** → **Inflammatory** → **Granulation tissue** → **Scar** (fibrosis).

Complications

Commonest	Haemorrhage — occult, or massive and life-threatening
Others	Pyloric obstruction (gastric outlet obstruction) · perforation and penetration · combined GU and DU
Rare	Malignant transformation of a benign ulcer — very rare, and more likely in GU than DU

P Must-remember pearls

- **Erosion stops above the muscularis mucosa; ulceration goes below it.** That one line separates gastritis from ulcer.
- **Curling = burns, Cushing = cranium, stress ulcer = shock/sepsis/trauma.**
- **Only autoimmune gastritis causes achlorhydria and pernicious anaemia** — H. pylori does not, because parietal cell damage is less severe.
- **Autoimmune gastritis is under 10% of chronic gastritis but the commonest cause of gastric atrophy.**
- **DU are 4× commoner than GU**, sit on the anterior duodenal wall, and H. pylori is present in almost all.
- **Haemorrhage is the commonest complication of PUD.** Malignant transformation is very rare, and favours GU over DU.

Q Test yourself

1. A patient with extensive burns develops upper GI bleeding. Endoscopy shows multiple small acute gastric ulcers. Which eponym applies?

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|------------------|------------------|
| A. Cushing ulcer | B. Curling ulcer |
| C. Stress ulcer | D. Meckel ulcer |

2. A 60-year-old has megaloblastic anaemia and achlorhydria. Gastric biopsy shows atrophy without organisms. Which mechanism explains her anaemia?

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| A. H. pylori-induced hypochlorhydria | B. Antibodies against parietal cells with loss of intrinsic factor |
| C. Chronic NSAID use | D. Chemical reactive gastritis |

3. A gastric biopsy shows tissue loss extending below the muscularis mucosa. Which term correctly describes this?

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|--------------------------|---------------|
| A. Erosion | B. Ulceration |
| C. Intestinal metaplasia | D. Atrophy |

4. Which statement about peptic ulcers is correct?

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| A. Gastric ulcers are four times commoner than duodenal | B. Duodenal ulcers usually arise on the posterior wall |
| C. Duodenal ulcers are four times commoner than gastric | D. Malignant transformation is commoner in DU than GU |

5. H. pylori infection of the antrum increases acid production. Which consequence follows most directly?

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| A. Pernicious anaemia | B. Increased risk of duodenal ulcer |
| C. Achlorhydria | D. Loss of intrinsic factor |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • B — Curling ulcers follow severe burns. Cushing ulcers accompany intracranial disease; generic stress ulcers follow shock, sepsis or trauma.

2 • B — Autoimmune gastritis destroys parietal cells, removing both HCl (achlorhydria) and intrinsic factor, so ileal B12 absorption fails — pernicious anaemia.

3 • B — By definition, ulceration extends below the muscularis mucosa; erosion is loss of superficial epithelium above it.

4 • C — DU are four times commoner, arise on the anterior duodenal wall near the pylorus, and malignant change favours GU.

5 • B — Antral H. pylori gastritis raises acid production, which increases duodenal ulcer risk. The atrophic/metaplastic route leads instead toward adenocarcinoma.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL1, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.